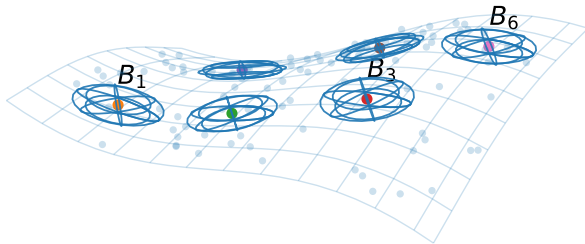


Normality Bubble Diffusion

Full MVTec AD results and model review



Huynh Thanh Dat 14 September 2026

Baselines → Baseline results → NBD → Comparison

One-class detection

Train on normal data

$$\mathcal{D}_N = \{x_i\}_{i=1}^n.$$

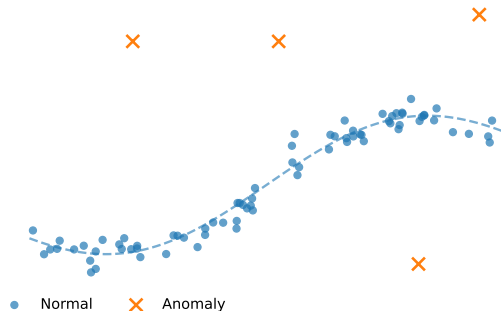
Score new observations

$$s(x) : \mathcal{X} \rightarrow \mathbb{R}.$$

A larger score means more anomalous.

Training and calibration: normal only.

Test: normal + anomaly.



Schematic: normal support can be curved.

Image benchmark: $z_p = f_{\text{CNN}}(x)_p$; the same frozen CNN descriptors are used by every method.

The original methods do not share one backbone

PatchCore — Roth et al.

ImageNet-pretrained Wide ResNet-50.

Freeze the CNN; extract intermediate patches.
Build a normal-feature memory, not a learned head.

GitHub: [amazon-science/patchcore-inspection](https://github.com/amazon-science/patchcore-inspection)

Deep SVDD — Ruff et al.

Trainable LeNet-type image CNN.

Initialize from a normal-image autoencoder;
then optimize the encoder with the SVDD loss.

GitHub: [lukasruff/Deep-SVDD](https://github.com/lukasruff/Deep-SVDD) (paper code)

SVDD — Tax and Duin

No CNN in the original method.

A kernel maps the chosen input features.
A quadratic program fits the enclosing sphere.

GitHub: [DMJTax/dd_tools](https://github.com/DMJTax/dd_tools) (author toolbox)

DROCC — Goyal et al.

Trainable LeNet on CIFAR-10.

Search for negatives in the image input space.
ImageNet-10 uses a different CNN: MobileNetV2.

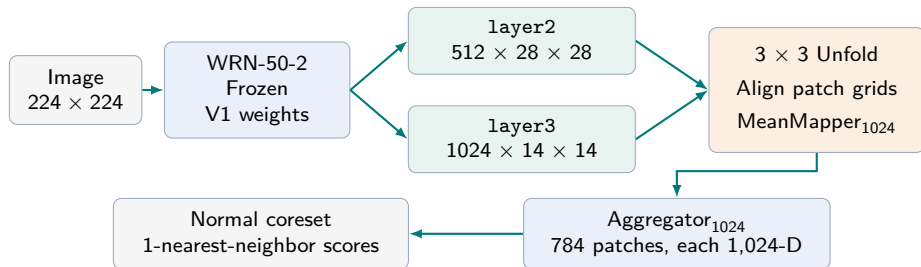
GitHub: [microsoft/EdgeML](https://github.com/microsoft/EdgeML) / DROCC

PatchScore is our project baseline. The source paper is **PatchCore**.

The existing MVTec tables below compare shared-CNN adaptations, not native-paper reproductions.

PatchCore: author feature extractor

Author Quick Guide: ImageNet-pretrained WRN-50 / 224 input / 1,024-D / 10% coreset.

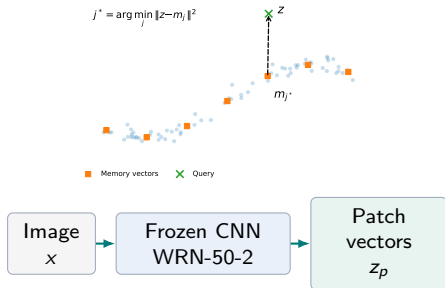


Our recorded v2 run: spatial averaging + concatenation gives **1,536-D**, not the author-code 1,024-D descriptor. Its top-1% mean is also different from the author-code image maximum.

Paper Eq. (7): extra neighbor reweighting. Inspected author code: maximum patch score. Separate targets.

GitHub: [patchcore.py](#) [common.py](#) [Author run command](#)

PatchScore: the recorded shared-CNN baseline



$$s_{PS}(z) = \min_{m_j \in \mathcal{M}} \|z - m_j\|_2^2.$$

Backbone used in this benchmark

Wide ResNet-50-2, frozen/eval.

ImageNet-1K V1; layer2 + layer3.

Resize 256; center crop 224.

Local 3×3 averaging, bilinear alignment.

28×28 patches; 1,536-D each. No global PCA.

Memory for the baseline comparison

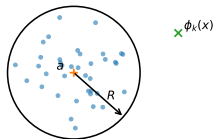
128 normal centers. Select farthest-first anchors from fit patches; take the mean of each anchor's 128-neighbor set.

Score a query by its nearest center. A larger-memory control is compared later.

All methods use the **mean of the top 1% patch scores** for each image.

SVDD and Deep SVDD on frozen CNN features

RBF SVDD



$$\min_{a, R, \xi \geq 0} R^2 + C \sum_i \xi_i,$$

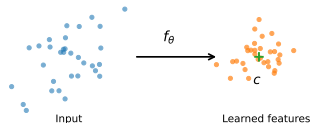
$$\text{s.t. } \|\phi_k(z_i) - a\|^2 \leq R^2 + \xi_i.$$

$$s_{\text{SVDD}}(z) = \|\phi_k(z) - a\|^2 - R^2.$$

Dual QP on 2,048 normal coreset patches.

$\nu = 0.1$; bandwidth from median pair distance.

Deep SVDD-head



$$\min_{\theta} \frac{1}{n} \sum_i \|g_\theta(z_i) - c\|^2 + \frac{\lambda}{2} \|\theta\|^2.$$

$$s_{\text{DSVDD}}(z) = \|g_\theta(z) - c\|^2.$$

Bias-free 1536 $\rightarrow$ 64 $\rightarrow$ 16 head; fixed $c \neq 0$.

50 AE + 100 SVDD epochs; all fit patches.

Author sources: [SVDD GitHub](#) [Deep SVDD paper-code GitHub](#) [Author PyTorch GitHub](#).

OCC / mvtec-full-v2 @ 91223d6 | Recorded adaptation: capped kernel support; the Deep SVDD head does not fine-tune the CNN.

Deep SVDD: collapse and safeguards

Shared-CNN setup: z is a frozen descriptor; only g_θ is trained.

Collapse $\neq$ compactness

Normal compactness is intended. A constant mapping loses all input information.

Trivial mapping

$$g_\theta(z) \equiv c \implies s(z) = 0.$$

Normal and anomalous queries get the same score.

A bias provides an easy constant-map solution:

$$g_\theta(z) = Wh(z) + b, \quad W = 0, \quad b = c.$$

Only the data term is necessarily zero.

AE gives an initialization, not a guarantee against collapse. The code logs final output variance as a diagnostic.

Controls in the current code

No bias. Both trainable head layers use `bias=False`.

Fixed nonzero center. Compute c once from normal outputs; do not optimize it.

ReLU. Avoid bounded activations with nonzero saturation.

AE pretraining. Reconstruct normal features, then discard the decoder before SVDD.

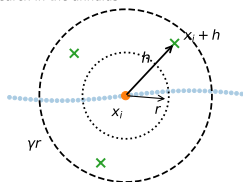
Ruff et al., Sec. 3.3. [Paper-code](#) [GitHub](#). Controls shown here refer to the recorded shared-feature head.

DROCC-head: negatives in CNN feature space

Normal label: 1 Generated label: 0

$$\min_{\theta} \frac{1}{n} \sum_i \left[\ell(g_{\theta}(z_i), 1) + \mu \max_{r \leq \|h\|_2 \leq \gamma r} \ell(g_{\theta}(z_i + h), 0) \right] + \frac{\lambda}{2} \|\theta\|^2.$$

Search in the annulus



Training

Warm up on normal data.

Ascend in h ; project back to the annulus.

Update θ to separate the two labels.

Settings used

1536 $\rightarrow$ 64 $\rightarrow$ 1 MLP; 100 epochs (10 warm-up).

50 ascent steps; $\gamma = 2$, $\mu = 1$.

$r = 0.5 \times$ median normal 5-NN distance.

Ascent step $0.05r$; score $s(z) = -g_{\theta}(z)$.

Native reference: [microsoft/EdgeML](#) / [DROCC \(GitHub\)](#).

OCC / mvtec-full-v2 @ 91223d6 | Recorded feature-space adaptation, not original image DROCC. Batch 8,192; all fit patches per epoch.

METHODS

MVTec AD: shared evaluation setup

	Train normal	Test normal	Test anomaly
All 15 categories	3,629	467	1,258

Fit: 60%
2,174 normal images

Reserved normal: 20%
721 images

Threshold: remainder
734 normal images

Per-seed totals across categories. Split by image, before extracting patches.

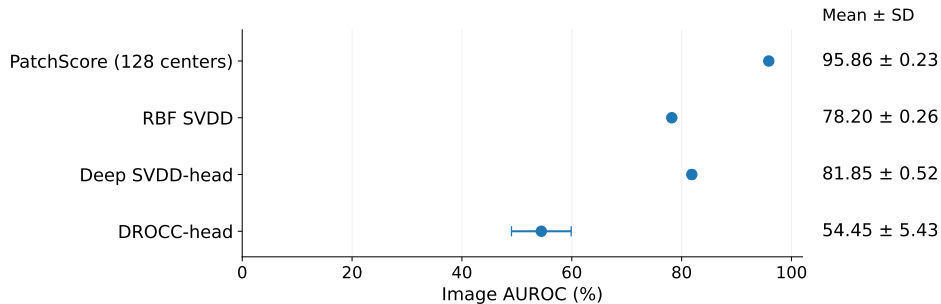
Reserved images are unused by these four baselines. Fit on the left subset; set image thresholds on the right subset.

Same frozen CNN descriptors and splits; seeds **0, 1, 2**. All test images retained.

Models and thresholds are fixed before testing.

Metrics: image AUROC / AP and FPR / TPR. Image score: mean of the top 1% patches.

Baseline results on MVTec AD



PatchScore with 128 centers has the highest image AUROC among these four baselines.

Mean $\pm$ sample SD of three seed means. Each seed averages the 15 categories equally.

OCC / mvtec-full-v2 @ 91223d6 | `summary.csv`. PatchScore (128 centers) is the published `PatchScore_same_centers` row.

Image thresholds and baseline operating points

One threshold per method, category and seed

Sort the n image scores in the normal threshold subset:

$$k = \lceil (n+1)(1-\alpha) \rceil, \quad \alpha = 0.05,$$

$$\tau = \begin{cases} s_{(k)}, & k \leq n, \\ +\infty, & k > n, \end{cases} \quad \hat{y} = \mathbf{1}\{s(x) > \tau\}.$$

No fit images or test labels are used to estimate τ .

Toothbrush: $n = 12$, so $k = 13$ and $\tau = +\infty$. Its zero FPR/TPR is retained.

$\alpha = 5\%$ is a calibration target, **not** the observed test FPR.

Measured test FPR and TPR (%)

Method	FPR	TPR
PatchScore (128 centers)	11.36	79.28
RBF SVDD	6.86	45.26
Deep SVDD-head	6.88	49.56
DROCC-head	3.51	8.38

Macro averages over categories and seeds. AUROC measures ranking; this table uses the fixed thresholds.

OCC / mvtec-full-v2 @ 91223d6 | summary.csv, reported threshold protocol. These image-threshold rules are shared by all methods.

Training diagnostics: Deep SVDD and DROCC

Deep SVDD-head

Image AUROC: **81.85** $\pm$ **0.52%**.

Output variance ratio

$$\frac{\text{final normal output variance}}{\text{initial normal output variance}}$$

Median: **0.0009754**.

Range: 0.0001308–0.0116951 over 45 runs.

Strong contraction. This alone does not establish a constant mapping or an implementation error.

Every fit patch was used in every epoch; batch size 8,192.

Terminal checkpoints were used regardless of their test performance.

DROCC-head

Image AUROC: **54.45** $\pm$ **5.43%**.

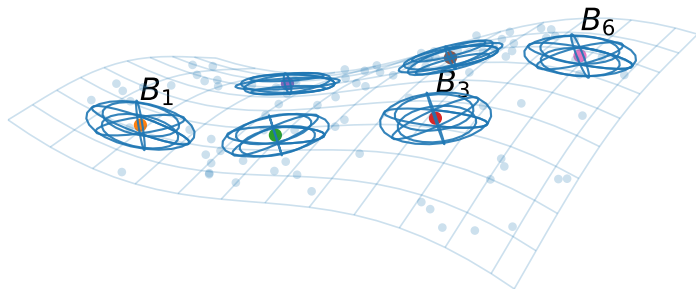
Final training loss

Median: **1.3862944** over 45 runs.

Measured test TPR: **8.38%**.

Weak detection in this shared-feature adaptation. This is not the original DROCC paper's benchmark result.

NBD: local bubbles, one shared graph



Schematic: local ellipsoids around a curved surface.

Each bubble stores

A center, tangent variation, and normal thickness.

The graph connects them

Diffusion compares the regions each bubble can reach.

Normal features $\rightarrow$ bubbles $\rightarrow$ shared graph $\rightarrow$ diffusion distance $\rightarrow$ query score

Hypothesis tested: a normal query has local support and attaches to a coherent region of the graph.

A local normality bubble

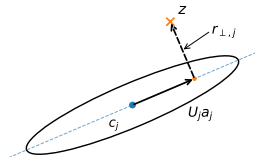
$$B_j = (c_j, U_j, \Lambda_j, \sigma_{\perp,j}^2), \quad U_j^\top U_j = I_{d_j}, \quad \Pi_j = U_j U_j^\top.$$

Local model

$$z = c_j + U_j a + \varepsilon_{\perp,j}, \quad a \perp \varepsilon_{\perp,j}, \\ a \sim \mathcal{N}(0, \Lambda_j), \quad \varepsilon_{\perp,j} \sim \mathcal{N}(0, \sigma_{\perp,j}^2 (I - \Pi_j)).$$

$$\Sigma_j = U_j \Lambda_j U_j^\top + \sigma_{\perp,j}^2 (I - \Pi_j).$$

Tangent motion + normal residual



Fit by local PCA

Neighborhood mean $\rightarrow c_j$; top d_j eigenpairs $\rightarrow U_j, \Lambda_j$.

$$\sigma_{\perp,j}^2 = \frac{1}{D - d_j} \sum_{k=d_j+1}^D \hat{\lambda}_{jk}.$$

Executed: 128 bubbles; 128 neighbors; $d_j = 16$. The same-center control uses the fitted neighborhood means c_j .

Scoring a query

$$a_j(z) = U_j^\top (z - c_j), \quad r_{\perp,j}(z) = (I - \Pi_j)(z - c_j).$$

Local energy

$$E_j(z) = \underbrace{\frac{a_j^\top (\Lambda_j + \epsilon I)^{-1} a_j}{d_j}}_{\text{tangent rarity}} + \beta \underbrace{\frac{\|r_{\perp,j}\|_2^2}{(D - d_j)(\sigma_{\perp,j}^2 + \epsilon)}}_{\text{off-manifold residual}}.$$

Absolute fit

$$S_B(z) = \min_{j \in \mathcal{C}(z)} E_j(z),$$

$$S_A(z) = -T_b \log \left[\frac{1}{K} \sum_{j \in \mathcal{C}(z)} e^{-E_j(z)/T_b} \right].$$

Attachment to the graph

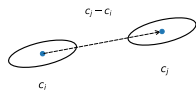
$$b_j(z) = \frac{e^{-E_j(z)/T_b}}{\sum_{k \in \mathcal{C}(z)} e^{-E_k(z)/T_b}}.$$

$K = |\mathcal{C}(z)|$ candidates; $b_j = 0$ outside.
 $\sum_j b_j(z) = 1.$

Executed: $K = 8$, $T_b = 1$, $\beta = 1$, $\epsilon = 10^{-6}$. This dimension-normalized energy is not Gaussian NLL.

Comparing two bubbles

Center displacement



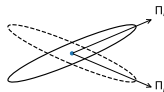
Earlier: Euclidean distance

$$r_c^E(i, j) = \|c_i - c_j\|_2^2.$$

Used here: two-way bubble energy

$$r_c^G(i, j) = \frac{1}{2} [E_i(c_j) + E_j(c_i)].$$

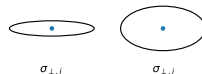
Frame disagreement



$$\begin{aligned} r_F(i, j) &= \|\Pi_i - \Pi_j\|_F^2 \\ &= d_i + d_j - 2 \|U_i^\top U_j\|_F^2. \end{aligned}$$

Compare tangent subspaces, not basis-vector signs.

Optional: thickness



$$r_\sigma(i, j) = \left[\log \frac{\sigma_{\perp, i}^2 + \epsilon}{\sigma_{\perp, j}^2 + \epsilon} \right]^2.$$

Not used in the reported benchmark.
Not a full covariance distance.

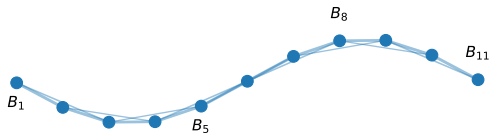
Executed graph: two-way center energy + frame disagreement. Thickness weight is zero.

The earlier Euclidean distance and the optional thickness term are shown for context, not as tested ablations.

From bubbles to a graph

Edge dissimilarity, before diffusion

$$\delta_B(i, j) = \frac{r_c(i, j)}{\tau_c} + \lambda_F \frac{r_F(i, j)}{\tau_F}.$$



Toy graph. Each node is one bubble; thicker edges have larger affinity.

Affinity and transition

For symmetric local candidate edges $\mathcal{E}$,

$$W_{ij} = \mathbf{1}_{\{(i,j) \in \mathcal{E}\}} e^{-\delta_B(i,j)} \quad (i \neq j),$$

$$W_{ii} = \eta > 0, \quad s_i = \sum_j W_{ij},$$

$$P_{ij} = \frac{W_{ij}}{s_i}, \quad \pi_i = \frac{s_i}{\sum_m s_m}.$$

Fit the graph once on normal data. Every query uses the same bubble nodes.

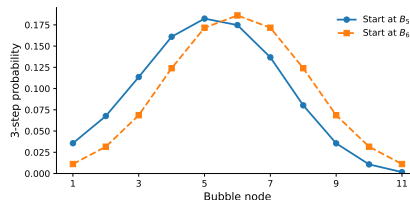
Executed: 12 graph neighbors; $\lambda_F = 1$, $\eta = 1$. No thickness penalty. δ_B need not be a metric.

NBD MODEL

Diffusion distance

Similar t -step transition distributions give a smaller diffusion distance:

$$D_t^2(i, j) = \sum_m \frac{[(P^t)_{im} - (P^t)_{jm}]^2}{\pi_m}.$$



Compute with eigenvectors

$$P\psi_\ell = \lambda_\ell \psi_\ell, \quad \sum_i \pi_i \psi_\ell(i) \psi_k(i) = \delta_{\ell k}.$$

$$D_t^2(i, j) = \sum_{\ell=1}^{M-1} \lambda_\ell^{2t} [\psi_\ell(i) - \psi_\ell(j)]^2.$$

$$\Phi_t(B_j) = (\lambda_1^t \psi_1(j), \dots, \lambda_L^t \psi_L(j)),$$

$$D_{t,L}^2(i, j) = \|\Phi_t(B_i) - \Phi_t(B_j)\|^2.$$

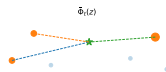
[4] Coifman–Lafon (2006). Executed: all $L = M - 1$ nonconstant modes, $\mathcal{T} = \{1, 3, 5\}$. The figure is schematic.

Does the query attach to one region?



Nearby bubbles: lower dispersion

Schematic diffusion coordinates. Circle area: $b_j(z)$. Star: weighted mean.



Distant bubbles: higher dispersion

$$S_D(z) = \frac{1}{2|\mathcal{T}|} \sum_{t \in \mathcal{T}} \sum_{i,j} b_i(z) b_j(z) D_{t,L}^2(i,j).$$

Equivalent variance at time t

$$\bar{\Phi}_t(z) = \sum_j b_j(z) \Phi_t(B_j),$$

$$S_{D,t}(z) = \sum_j b_j(z) \left\| \Phi_t(B_j) - \bar{\Phi}_t(z) \right\|^2.$$

Frame inconsistency

$$S_F(z) = \frac{1}{2} \sum_{i,j} b_i(z) b_j(z) r_F(i,j).$$

Executed: $\mathcal{T} = \{1, 3, 5\}$.

A one-bubble outlier can have $S_D \approx 0$; S_B and S_A still measure its absolute mismatch.

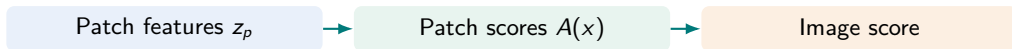
NBD: component scaling and final score

Local fit + diffusion and frame consistency

$$S_{\text{NBD}}(z) = \tilde{S}_B(z) + \tilde{S}_A(z) + \tilde{S}_D(z) + \lambda_F \tilde{S}_F(z).$$

NBD only: $\mathcal{Z}_{\text{cal}}$ contains patches from the 721 reserved normal images:

$$\tilde{S}_q(z) = -\log \left[\frac{1 + \sum_{u \in \mathcal{Z}_{\text{cal}}} \mathbf{1}_{\{S_q(u) \geq S_q(z)\}}}{|\mathcal{Z}_{\text{cal}}| + 1} \right], \quad q \in \{B, A, D, F\}.$$



$$A(x)_p = S_{\text{NBD}}(z_p).$$

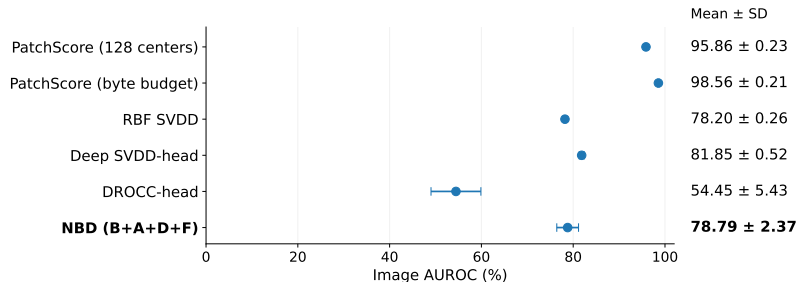
$$S_{\text{image}}(x) = \frac{1}{|\mathcal{P}_\rho(x)|} \sum_{p \in \mathcal{P}_\rho(x)} A(x)_p.$$

$\mathcal{P}_\rho$: top 1% (8 of 784 patches). $\lambda_F = 1$. This is not a calibrated anomaly probability.

Component scaling uses the reserved subset, not the 734 threshold images.

After aggregation, NBD uses the same image-threshold rule as the baselines.

NBD versus the baseline methods



The 128-center PatchScore row is unchanged. The extra byte-budget control stores normal patches up to the NBD geometry/graph array budget; CNN and calibration state are excluded.

NBD minus 128 centers: -17.07 ± 2.22 pp.

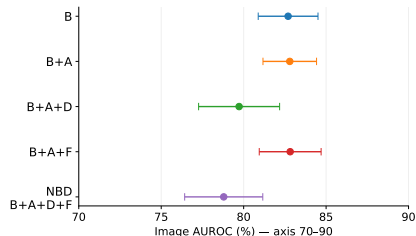
NBD minus byte budget: -19.77 ± 2.29 pp.

OCC / mvtec-full-v2 @ 91223d6 | [summary.csv](#), [paired_deltas.csv](#). Same data, seeds, CNN and top-1% aggregation. NBD is $B + A + D + F$.

NBD COMPARISON

Ablation: diffusion reduced AUROC in this configuration

Local and graph score variants



B: best bubble; *A*: soft support;
D: diffusion dispersion; *F*: frame inconsistency.

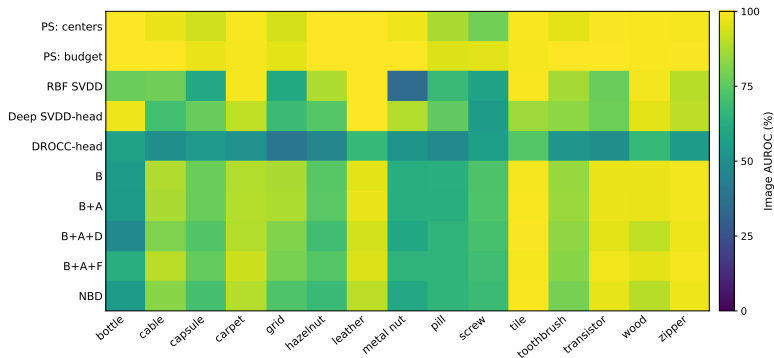
Paired changes (AUROC pp)

Change	Mean $\pm$ SD
Add <i>A</i> to <i>B</i>	+0.10 $\pm$ 0.20
Add <i>D</i> to <i>B + A</i>	-3.07 $\pm$ 0.92
Add <i>F</i> to <i>B + A</i>	+0.03 $\pm$ 0.61
Add <i>D</i> to <i>B + A + F</i>	-4.03 $\pm$ 0.73
Add <i>F</i> to <i>B + A + D</i>	-0.93 $\pm$ 0.33

B + A + F: $82.82 \pm 1.88\%$. Declared NBD: $78.79 \pm 2.37\%$.

The better ablation remains an ablation, not the final NBD result.

All 15 categories, all 10 variants



NBD ranges from 54.84% on bottle to 99.17% on tile.

The full color scale is 0–100%; results below 50% remain visible.

OCC / mvtec-full-v2 @ 91223d6 | category_summary.csv. Each cell is the mean of three seeds; exact values are included in the CSV.

Appendix: complete measured results

Method	Image AUROC (%)	AP (%)	FPR (%)	TPR (%)
PatchScore (128 centers)	95.86 ± 0.23	98.67 ± 0.05	11.36	79.28
PatchScore (byte budget)	98.56 ± 0.21	99.59 ± 0.06	13.23	87.04
RBF SVDD	78.20 ± 0.26	90.46 ± 0.17	6.86	45.26
Deep SVDD-head	81.85 ± 0.52	91.58 ± 0.32	6.88	49.56
DROCC-head	54.45 ± 5.43	75.73 ± 3.29	3.51	8.38
Bubble B	82.70 ± 1.81	93.19 ± 0.85	11.78	54.04
Bubble $B+A$	82.80 ± 1.62	93.26 ± 0.75	11.07	53.80
Bubble $B+A+D$	79.72 ± 2.46	91.53 ± 1.29	10.31	47.72
Bubble $B+A+F$	82.82 ± 1.88	93.27 ± 0.91	10.89	53.20
NBD ($B+A+D+F$)	78.79 ± 2.37	91.06 ± 1.44	10.81	47.27

AUROC/AP: category-macro mean $\pm$ sample SD across three seeds. FPR/TPR: macro means at normal-only thresholds.

NBD at its normal-only threshold: FPR 10.81%, TPR 47.27%. No test-optimized cutoffs.

Method sources

[1] Tax and Duin (2004).

Support Vector Data Description.

[2] Ruff et al. (2018).

Deep One-Class Classification.

[3] Goyal et al. (2020).

DROCC: Deep Robust One-Class Classification.

[4] Coifman and Lafon (2006).

Diffusion maps.

PatchCore reference implementation MVTec AD

Executed benchmark

dathuynh1108/OCC @ 91223d6

results/mvtec-full-v2/

REPORT.md, summary.csv

category_summary.csv

paired_deltas.csv

SLIDE_UPDATE_HANDOFF.md

Locked settings: configs/full.json.

Backbone: WRN-50-2, ImageNet-1K V1.

This slide package

review.tex, assets/,

CSV snapshots and plotting code.

Reported audit: 450 metric rows; 51,750

image predictions. Replay: 135 groups; max error 0.0.

OCC / mvtec-full-v2 @ 91223d6 | Snapshot: 14 September 2026. No benchmark was rerun for this editorial update.

GitHub sources and rerun scope

Method	Clickable GitHub source
PatchCore	amazon-science/patchcore-inspection fcaa92f
SVDD	DMJTax/dd_tools efaaf04 (author MATLAB reference)
Deep SVDD	lukasruff/Deep-SVDD e20f18c (original paper code)
	lukasruff/Deep-SVDD-PyTorch 1901612 (author port)
DROCC	microsoft/EdgeML / examples/pytorch/DROCC 81025fc
Our results	dathuynh1108/OCC / results/mvtec-full-v2 91223d6

A. Native-source reproduction

Use each paper's data and training.
Keep paper/code discrepancies explicit.

B. Controlled NBD rerun

Share the author PatchCore encoder.
Label changed methods as adaptations.

Source audit and run instructions: `handoff/CODEX_REPRODUCE.md`.

Native-source reruns are **pending**. Existing numbers remain the measured v2 results.